

Field Knowledge Objects for Trades

Below is a collection of mentor-worthy knowledge objects gathered from Reddit and trade forums. Each entry captures an experienced tradesperson's real-world lesson in a structured format for Jack's knowledge base. The goal is to preserve practical tips, common mistakes, root causes, and solutions that can help apprentices work safer and smarter. Citations reference the original source lines for verification.

Welding

1. Drag the rod to avoid slag inclusion

2. **Trade:** Welding

3. **Specialty:** SMAW (6013 vertical-up)

4. **Scenario:** Student welding vertical up with 6013 rod sees slag trapped in bead.

5. **Problem:** Slag inclusion and a messy weld when traveling upward.

6. **Root Cause:** Improper technique—pushing and pausing in the puddle causes slag to roll ahead.

7. **Solution:** Use a slight drag angle and stay on the leading edge of the puddle; maintain consistent speed ¹.

8. **Field Tip:** Think of dragging 6013 as more of a drag than push; let the slag follow behind the arc ².

9. **Common Mistake:** Mashing up the puddle by lingering too long or “weaving” aggressively ¹.

10. **Safety Note:** Maintain proper PPE and shield—repeated restarts increase spatter.

11. **When Not to Use:** N/A (drag technique applies to most 6013 vertical-up welds).

12. **Confidence:** 7/10

13. **Tags:** welding, slag inclusion, vertical up, 6013

14. **Source Link:** ¹

15. **Supporting Quote:** “Slag inclusion is a result of improper angle and speed. Drag it and stay on the leading edge of the puddle” ¹.

16. Drag to prevent puddle mashing

17. **Trade:** Welding

18. **Specialty:** SMAW (6013 vertical-up)

19. **Scenario:** Welds turn to a glob of slag at the start of the weld.

20. **Problem:** The puddle builds up and traps slag.

21. **Root Cause:** Not dragging the electrode; moving too slowly off the arc start ².

22. **Solution:** Drag the 6013 rod so the puddle moves away from the slag; maintain a steady pace ².

23. **Field Tip:** Practice starting quickly and letting the puddle flow; if the rod sticks, increase amperage slightly.

24. **Common Mistake:** Starting too slow and letting the puddle mash up; pushing instead of dragging ².

25. **Safety Note:** Avoid excessive amperage that could burn through thin material.
26. **When Not to Use:** On downhill SMAW runs where a slight push is needed.
27. **Confidence:** 6/10
28. **Tags:** welding, slag inclusion, vertical up, drag technique
29. **Source Link:** [2](#)
30. **Supporting Quote:** "I'm not dragging 6013 and I'm too slow off the arc start... the puddle mashes up" [2](#).
31. **Touch-lift restart for 7018 electrodes**
32. **Trade:** Welding
33. **Specialty:** SMAW (7018 restarts)
34. **Scenario:** Difficulty restarting 7018 electrodes on a vertical weld.
35. **Problem:** Porosity and arc start issues when restarting mid-bead.
36. **Root Cause:** Arc not established cleanly and amperage set too low or high.
37. **Solution:** Use a touch-lift technique—strike, then lift off to create a brief "white arc," adjust amperage for 1–3 globules, then weave over previous ripple [3](#).
38. **Field Tip:** Practice on scrap; dial heat so you see globules before removing slag [3](#).
39. **Common Mistake:** Holding electrode too long at restart, overheating the rod, or not cleaning slag before restarting.
40. **Safety Note:** Hot starts can spatter; wear proper clothing.
41. **When Not to Use:** On downhill 7018 applications where continuous travel prevents restarts.
42. **Confidence:** 8/10
43. **Tags:** welding, 7018, restarts, touch lift
44. **Source Link:** [3](#)
45. **Supporting Quote:** "Touch and lift... establish the white arc... adjust heat until there are 1–3 globules" [3](#).
46. **U-turn tie-in on 7018 welds**
47. **Trade:** Welding
48. **Specialty:** SMAW (7018 restarts)
49. **Scenario:** Joining a restart into the existing bead.
50. **Problem:** Tie-ins are visible and undercut.
51. **Root Cause:** Improper motion when blending new bead into old bead.
52. **Solution:** Think of the tie-in like a quick U-turn; weave into the existing bead then return to the pattern quickly [4](#).
53. **Field Tip:** Reduce travel speed as you approach the tie-in; pause slightly on the toes to fill the crater.
54. **Common Mistake:** Dwelling too long in the centre and creating a hump at the restart [4](#).
55. **Safety Note:** Always remove slag and clean the area before a restart.
56. **When Not to Use:** N/A—works for most 7018 tie-ins.
57. **Confidence:** 7/10
58. **Tags:** welding, tie-in, 7018, restart
59. **Source Link:** [4](#)

60. **Supporting Quote:** “Think of it like a quick U-turn back into your weave pattern” ⁴ .

61. Spider-Man pattern for vertical MIG

62. **Trade:** Welding

63. **Specialty:** GMAW (vertical up)

64. **Scenario:** Inconsistent penetration and poor bead appearance on vertical MIG.

65. **Problem:** Spending too long in the centre of the weld and not filling the toes properly.

66. **Root Cause:** Circular motion cuts into sides without allowing puddle to push into toes ⁵ .

67. **Solution:** Use a “Spider-Man” pattern—slow pass at bottom, then swing across top of previous pass; limit time in centre; adjust wire feed (120–130 WFS) and voltage (18.5–19.4 V) ⁵ .

68. **Field Tip:** Think of swinging like Spider-Man: subtle swoop, jump to the other side; keep travel speed consistent ⁵ .

69. **Common Mistake:** Using a simple circular pattern and staying too long in the puddle.

70. **Safety Note:** Maintain travel speed to avoid burn-through on thin sections.

71. **When Not to Use:** Very thick material where stringers are preferred.

72. **Confidence:** 6/10

73. **Tags:** welding, MIG, vertical up, pattern

74. **Source Link:** ⁵

75. **Supporting Quote:** “Swing like Spider-Man... limit your time in the centre... start at 120–130 WFS and 18.5–19.4 V” ⁵ .

76. Inverted U pattern for vertical MIG

77. **Trade:** Welding

78. **Specialty:** GMAW (vertical up)

79. **Scenario:** Student learning to run vertical MIG on moderate thickness.

80. **Problem:** Poor bead shape and lack of fusion at toes.

81. **Root Cause:** Incorrect travel angle and pattern.

82. **Solution:** Hold a 10–15° up angle and run an inverted U pattern; pause on the toes for a two-count and move quickly across the centre ⁶ .

83. **Field Tip:** Move fast from toe to toe; avoid sweeping motions; keep arc length short ⁶ .

84. **Common Mistake:** Sweeping or weaving in a circular pattern; holding too long in the middle.

85. **Safety Note:** Ensure proper shielding gas flow when moving quickly.

86. **When Not to Use:** On thin gauge sheet metal where stringers are better.

87. **Confidence:** 6/10

88. **Tags:** welding, MIG, vertical up, inverted U

89. **Source Link:** ⁶

90. **Supporting Quote:** “10–15 degree up angle. Run an inverted U pattern and hold on the toes for a two-count. Move fast from toe to toe; don’t sweep” ⁶ .

91. Downhill welding where allowed

92. **Trade:** Welding

93. **Specialty:** GMAW and FCAW

94. **Scenario:** Fabricating bumper-to-frame welds; vertical up vs down debate.

95. **Problem:** Student struggles with vertical-up MIG when downhill could suffice.

96. **Root Cause:** Upward welding is more difficult and often unnecessary for non-structural work.

97. **Solution:** An instructor with over 25 years of pipeline experience advised that, unless it's mission-critical, weld down whenever you have space ⁷.

98. **Field Tip:** For MIG there is no slag; downhill is faster and easier; still watch for adequate penetration and avoid quenching ⁷.

99. **Common Mistake:** Welding vertical up on non-critical joints where downhill would suffice; resulting in poor appearance and unnecessary effort.

100. **Safety Note:** Do not use downhill technique on structural connections requiring specific procedures.

101. **When Not to Use:** On code-driven work or materials requiring vertical up (e.g., 7018 SMAW).

102. **Confidence:** 5/10

103. **Tags:** welding, downhill, vertical up, MIG

104. **Source Link:** ⁷

105. **Supporting Quote:** "If you have the space to weld up, you may as well weld down. It's a lot easier" ⁷.

106. **Clean everything for TIG**

107. **Trade:** Welding

108. **Specialty:** GTAW (TIG)

109. **Scenario:** TIG welds appear contaminated like a sparkler, despite changing gas and tungsten.

110. **Problem:** Porosity and contaminated TIG welds.

111. **Root Cause:** Dirty base metal, contaminated filler rods, or oily gloves ⁸.

112. **Solution:** Wire brush the joint, then wipe with acetone; use a clean brush and clean filler rods; dedicate gloves for TIG to avoid cross-contamination ⁸.

113. **Field Tip:** If your gloves are dirty from MIG or grinding, switch to clean TIG gloves before handling the filler ⁹.

114. **Common Mistake:** Relying solely on new gas or tungsten while ignoring preparation.

115. **Safety Note:** Acetone is flammable; keep away from ignition sources.

116. **When Not to Use:** N/A; cleaning is always beneficial.

117. **Confidence:** 8/10

118. **Tags:** welding, TIG, contamination, cleaning

119. **Source Link:** ⁸

120. **Supporting Quote:** "Wire brush first, then acetone... filler could be oily... I try to just use gloves for TIG and gloves for stick/mig" ⁹.

121. **Check for gas leaks in TIG setup**

122. **Trade:** Welding

123. **Specialty:** GTAW (TIG)
124. **Scenario:** TIG weld contamination persists after cleaning.
125. **Problem:** Shielding gas contamination causing porosity.
126. **Root Cause:** Air entering system through leaks, bad O-rings, or incorrect gas mix ¹⁰ .
127. **Solution:** Leak-check all connections from bottle to machine and torch with soapy water; feel for cold breeze near fittings; inspect the back-cap O-ring; verify bottle contains 100% argon ¹⁰ .
128. **Field Tip:** If contamination continues, weld a test bead on known clean material to isolate whether the gas or material is at fault ¹¹ .
129. **Common Mistake:** Assuming “bad gas” without checking for leaks or incorrect bottle mix ¹⁰ .
130. **Safety Note:** Always perform leak checks in well-ventilated areas to avoid asphyxiation.
131. **When Not to Use:** When using flux-shielded processes (SMAW or FCAW) that do not rely on external shielding gas.
132. **Confidence:** 7/10
133. **Tags:** welding, TIG, gas leak, shielding
134. **Source Link:** ¹⁰
135. **Supporting Quote:** “Leak check all your connections... use soapy water... confirm the gas is 100% argon... check the o-ring on the back cap” ¹² .

136. Assemble TIG torch correctly

137. **Trade:** Welding
138. **Specialty:** GTAW (TIG)
139. **Scenario:** Persistent contamination despite clean metal and good gas.
140. **Problem:** Argon leaking or swirling in the torch assembly.
141. **Root Cause:** Loose gas lens, misaligned collet, or stepping on gas hose ¹³ .
142. **Solution:** Install lens first, then cup; insert collet and tungsten; tighten back cap; ensure hose routing prevents stepping on it ¹³ .
143. **Field Tip:** Double-check torch build whenever changing tungsten or cups.
144. **Common Mistake:** Assembling components out of order or leaving gas lens loose.
145. **Safety Note:** Gas leaks can cause oxygen to displace air around the arc—ventilate properly.
146. **When Not to Use:** N/A.
147. **Confidence:** 6/10
148. **Tags:** welding, TIG, torch assembly, contamination
149. **Source Link:** ¹³
150. **Supporting Quote:** “Run your lens in tight first, then cup, pop your collet & tungsten in and screw down the cap” ¹³ .

151. Test piece to isolate TIG contamination

152. **Trade:** Welding
153. **Specialty:** GTAW (TIG)
154. **Scenario:** Cannot determine whether contamination is from gas or material.
155. **Problem:** Erratic arc and porosity on specific workpiece.
156. **Root Cause:** Base material may contain impurities or moisture.

157. **Solution:** Weld on another known-clean piece of metal; if contamination disappears, the original material is suspect; if it persists, equipment or gas is the culprit ¹¹ .
158. **Field Tip:** Increase gas flow 5–15 CFH temporarily to flush any stray air ¹¹ .
159. **Common Mistake:** Immediately blaming gas supplier without isolating variables.
160. **Safety Note:** Do not exceed recommended gas flow; high flow may create turbulence.
161. **When Not to Use:** When material cannot be substituted.
162. **Confidence:** 5/10
163. **Tags:** welding, TIG, troubleshooting, contamination
164. **Source Link:** ¹¹
165. **Supporting Quote:** “Could be contamination from the base material, filler or tungsten... try a bead on another piece of metal” ¹¹ .
166. **Dedicated gloves for TIG**
167. **Trade:** Welding
168. **Specialty:** GTAW (TIG)
169. **Scenario:** TIG welding contamination from dirty hands.
170. **Problem:** Oil or dirt from gloves contaminates filler rods and tungsten.
171. **Root Cause:** Using the same gloves for grinding, MIG and TIG transfers contaminants.
172. **Solution:** Keep separate gloves for TIG; ensure they are clean and oil-free ⁹ .
173. **Field Tip:** Clean gloves occasionally with mild detergent to remove accumulated oil.
174. **Common Mistake:** Wiping filler rods with dirty gloves just before welding ⁹ .
175. **Safety Note:** Leather gloves can absorb flammable solvents; keep solvents separate.
176. **When Not to Use:** N/A.
177. **Confidence:** 5/10
178. **Tags:** welding, TIG, gloves, contamination
179. **Source Link:** ⁹
180. **Supporting Quote:** “Use fresh brushes... filler could be oily... I try to just use gloves for TIG and gloves for stick/mig” ⁹ .

Electrical

1. **Use a door frame to square 90° bends**
2. **Trade:** Electrical
3. **Specialty:** Conduit bending (EMT)
4. **Scenario:** Measuring the accuracy of 90° conduit bends.
5. **Problem:** Floors or tables may not be square, giving false results.
6. **Root Cause:** Using untrusted surfaces as references.
7. **Solution:** Use a door frame to check 90° bends; frames are typically plumb and square.
8. **Field Tip:** Keep a small combination square in your pouch for quick checks.
9. **Common Mistake:** Checking bends against the floor and then discovering they are off when installing.
10. **Safety Note:** Be mindful of pinch points when pressing conduit against frames.

11. **When Not to Use:** When door frames themselves are not square (old buildings).
12. **Confidence:** 6/10
13. **Tags:** electrical, conduit bending, 90 degree, checking
14. **Source Link:**
15. **Supporting Quote:** "I use door frames to check my 90s... walls and floors aren't always square".
16. **Pre-bend with #14 solid wire**
17. **Trade:** Electrical
18. **Specialty:** Conduit layout (EMT)
19. **Scenario:** Laying out complex conduit runs with multiple bends.
20. **Problem:** Wasting conduit due to mis-bent runs.
21. **Root Cause:** Difficult to visualise compound offsets and saddles on rigid conduit.
22. **Solution:** Carry a piece of #14 solid copper ground wire; bend it to mimic the run; once satisfied, replicate bends on real conduit.
23. **Field Tip:** The pre-bent wire also helps measure distances accurately by holding it to the studs before bending.
24. **Common Mistake:** Attempting complex bends without a template, leading to multiple wasted pieces.
25. **Safety Note:** Use caution when handling bare copper wire around energized circuits.
26. **When Not to Use:** Simple single bends where direct measurement is faster.
27. **Confidence:** 7/10
28. **Tags:** electrical, conduit bending, pre-bend, layout
29. **Source Link:**
30. **Supporting Quote:** "Carry a piece of #14 solid ground wire... pre-bend and then replicate on your conduit".
31. **Use the iBend app for complex saddles**
32. **Trade:** Electrical
33. **Specialty:** Conduit bending
34. **Scenario:** Making four-point saddles and complicated bends.
35. **Problem:** Difficulty marking centers and offsets accurately.
36. **Root Cause:** Manual measurements and repositioning introduce error.
37. **Solution:** Use the iBend app to calculate and mark the center; bend-slide-bend technique avoids repositioning; ideal for four-point saddles.
38. **Field Tip:** Combine the app with your pencil and level; mark all bends before going to the bender.
39. **Common Mistake:** Attempting to eyeball complex bends without referencing the center.
40. **Safety Note:** Keep phone clean; avoid dropping it in the jobsite.
41. **When Not to Use:** Simple offsets or when phone usage is restricted on site.
42. **Confidence:** 6/10
43. **Tags:** electrical, conduit, iBend, saddle, four-point
44. **Source Link:**

45. **Supporting Quote:** “iBend calculates the center... just mark it, slide, bend and you’re done”.

46. **Multipliers for offsets**

47. **Trade:** Electrical

48. **Specialty:** Conduit bending

49. **Scenario:** Laying out offsets for conduit runs.

50. **Problem:** Calculating spacing between bends for various angles.

51. **Root Cause:** Relying solely on memory or guesswork.

52. **Solution:** Use multipliers: for 10° bends multiply the offset height by 6; for 30° by 2; for 45° by 1.4.

53. **Field Tip:** Write the multipliers on your bender or tool bag; use them consistently.

54. **Common Mistake:** Using wrong multiplier or mixing degrees, resulting in offsets that don’t line up..

55. **Safety Note:** Ensure you maintain proper bending posture to avoid injury.

56. **When Not to Use:** When precise geometric calculations are required (e.g., large-diameter conduit); then use trig functions.

57. **Confidence:** 7/10

58. **Tags:** electrical, conduit, offset multipliers

59. **Source Link:**

60. **Supporting Quote:** “10°: multiply height by 6, 30°: multiply height by 2, 45°: multiply height by 1.4”.

61. **Star-notch-arrow saddle bending technique**

62. **Trade:** Electrical

63. **Specialty:** Conduit bending

64. **Scenario:** Creating a three-bend saddle over a small obstacle.

65. **Problem:** Incorrect spacing and inconsistent saddle height.

66. **Root Cause:** Lack of a repeatable procedure.

67. **Solution:** Measure the obstacle height, double it for the centre mark; mark star, notch and arrow on the conduit; bend 22° on the star, 45° on the notch, and 22° on the arrow.

68. **Field Tip:** Keep bends on the same plane and avoid dog legs by rotating the pipe correctly..

69. **Common Mistake:** Forgetting to double the height, resulting in a saddle that is too shallow; misaligning marks..

70. **Safety Note:** Ensure your bender’s shoe is locked before applying force.

71. **When Not to Use:** N/A—standard technique for three-bend saddles.

72. **Confidence:** 6/10

73. **Tags:** electrical, conduit bending, saddle, star-notch-arrow

74. **Source Link:**

75. **Supporting Quote:** “Mark the centre by doubling the height... bend 22° on the star, 45° on the notch, 22° on the arrow”.

76. **Quick box offset trick**

77. **Trade:** Electrical

78. **Specialty:** Conduit bending
79. **Scenario:** Making a short box offset by eye.
80. **Problem:** Inconsistent offsets; over- or under-bending.
81. **Root Cause:** Overcomplicating small offsets with math ¹⁴ .
82. **Solution:** For box offsets, stick 1/4" of conduit into the bender shoe; make a slight bend; flip pipe 180°; insert to the same depth; pull until the first bend stops you and make another slight bend ¹⁴ .
83. **Field Tip:** Avoid measuring; this "touch and flip" method produces consistent offsets quickly ¹⁴ .
84. **Common Mistake:** Trying to apply long-offset math to small box offsets.
85. **Safety Note:** Mark one side of the conduit to avoid flipping incorrectly.
86. **When Not to Use:** Large offsets or those crossing obstructions.
87. **Confidence:** 8/10
88. **Tags:** electrical, conduit, box offset, trick
89. **Source Link:** ¹⁴
90. **Supporting Quote:** "Take the pipe... leave about a quarter of an inch sticking out... give it a slight bend... flip 180 degrees... slight bend again—works every time" ¹⁴ .

91. Mark opposite sides for 10° and 15° offsets

92. **Trade:** Electrical

93. **Specialty:** Conduit bending
94. **Scenario:** Creating small offsets without dog-legging.
95. **Problem:** Offsets come out crooked or misaligned.
96. **Root Cause:** Incorrect alignment and spacing ¹⁵ .
97. **Solution:** Mark the conduit on opposite sides; for a 10° offset, mark distances of 3" and 5"; for 15° offsets, mark 2" and 4"; place the first mark on the arrow and bend; rotate 180°, align second mark, then bend to the second angle ¹⁵ .
98. **Field Tip:** Always rotate the conduit exactly 180° between bends to prevent dog legs ¹⁵ .
99. **Common Mistake:** Mis-rotating or mis-spacing marks, leading to dog-legged offsets.
100. **Safety Note:** Use gloves when bending rigid conduit to prevent pinched hands.
101. **When Not to Use:** Offsets with larger angles (30° or 45°) where other formulas apply.
102. **Confidence:** 7/10
103. **Tags:** electrical, conduit, offset, 10 degree, 15 degree
104. **Source Link:** ¹⁵
105. **Supporting Quote:** "Mark your pipe at 2" and either 4" or 5"... rotate so the next mark is on top... bend 10° or 15° accordingly" ¹⁵ .

106. Look down both sides to avoid dog-legs

107. **Trade:** Electrical

108. **Specialty:** Conduit bending
109. **Scenario:** Ensuring a straight offset without twist.
110. **Problem:** Dog-leg (twisted) offsets that don't align with boxes or connectors.
111. **Root Cause:** Uneven rotation or misalignment between bends (implicit in commentary).

- 112. **Solution:** After completing bends, look down both sides of the pipe; if one side appears off while the other is straight, gently adjust by hand before installation. Ensure the bender handle is properly aligned when rotating.
- 113. **Field Tip:** Place a mark on the conduit to align with the bender handle; this helps maintain proper orientation.
- 114. **Common Mistake:** Trusting feel rather than visual confirmation; failing to adjust before install.
- 115. **Safety Note:** Wear safety glasses when looking down the pipe to avoid metal shavings.
- 116. **When Not to Use:** When using specialized bending equipment that prevents dog-legs automatically.
- 117. **Confidence:** 5/10
- 118. **Tags:** electrical, conduit, dog-leg, alignment
- 119. **Source Link:** (implicit summary from bending discussions)
- 120. **Supporting Quote:** Not applicable; compiled from field experience discussions.

Plumbing

1. Smooth pipe and joint compound for compression fittings

2. **Trade:** Plumbing

3. **Specialty:** Water supply compression fittings

4. **Scenario:** New compression joint leaks on copper supply line.

5. **Problem:** Drips at a newly installed compression fitting.

6. **Root Cause:** Pipe not smooth, ferrule not lubricated, or over-tightening ¹⁶ ¹⁷ .

7. **Solution:** Smooth the tubing with fine-grit sandpaper; apply jointing compound to the ferrule; snug nut, then tighten a half turn; if leak persists, replace the ferrule and nut ¹⁶ ¹⁷ .

8. **Field Tip:** Wrap thread tape around the olive only as a temporary fix; replacement is the proper solution ¹⁷ .

9. **Common Mistake:** Over-tightening the nut, which deforms the ferrule and causes leaks ¹⁷ .

10. **Safety Note:** Shut off water supply before disassembling the fitting.

11. **When Not to Use:** Compression fittings on gas lines without approved compound.

12. **Confidence:** 7/10

13. **Tags:** plumbing, compression fitting, leak repair

14. **Source Link:** ¹⁶ ¹⁸ ¹⁷

15. **Supporting Quote:** "Use fine grit sandpaper... apply joint compound on the ferrule... don't over tighten... sometimes you just need to replace the ferrule" ¹⁷ .

16. Thread tape around ferrule—temporary fix

17. **Trade:** Plumbing

18. **Specialty:** Compression fittings

19. **Scenario:** Homeowner attempts to fix minor compression leak.

20. **Problem:** Slight seeping at the nut after installation.

21. **Root Cause:** Imperfect ferrule or minor scratch on tubing ¹⁷ .

22. **Solution:** Wrap PTFE tape around the ferrule to slow the leak; ultimately replace the fitting when convenient ¹⁷ .

- 23. **Field Tip:** Do not rely on thread tape as a permanent solution; it is a stop-gap measure.
- 24. **Common Mistake:** Layering multiple wraps of tape instead of addressing the root cause.
- 25. **Safety Note:** PTFE tape is for water; do not use on gas compression fittings.
- 26. **When Not to Use:** On new installations—fix underlying issue instead.
- 27. **Confidence:** 4/10
- 28. **Tags:** plumbing, compression fitting, PTFE tape
- 29. **Source Link:** [17](#)
- 30. **Supporting Quote:** “I’ve seen people wrap the ferrule with thread tape but replacing the fitting is better” [17](#) .

31. Re-crimp PEX rings to stop leaks

- 32. **Trade:** Plumbing
- 33. **Specialty:** PEX
- 34. **Scenario:** Leak at PEX connection near a circulation pump.
- 35. **Problem:** Water seeping around crimp ring.
- 36. **Root Cause:** Loose or improperly crimped PEX ring or ridges on fitting [19](#) .
- 37. **Solution:** Wiggle the ring to see if it moves; if loose, re-crimp with proper tool; if tight but leaking, cut off ring, inspect fitting for ridges, and install a new ring; replace the fitting if damaged [19](#) [20](#) .
- 38. **Field Tip:** Use a PEX-compatible reducer or coupler instead of reusing suspect fittings [20](#) .
- 39. **Common Mistake:** Assuming the pump flange is leaking when the ring is the culprit [20](#) .
- 40. **Safety Note:** Turn off water and depressurize the system before cutting the ring.
- 41. **When Not to Use:** On expansion PEX (uses expansion rings instead of crimp).
- 42. **Confidence:** 7/10
- 43. **Tags:** plumbing, PEX, leak repair, crimp ring
- 44. **Source Link:** [19](#) [20](#)
- 45. **Supporting Quote:** “Wiggle the ring... if loose re-crimp; if still leaking cut off the ring and put on a new one... inspect the fitting for ridges” [19](#) [20](#) .

46. Use 45s or 22s for steep drain drops

- 47. **Trade:** Plumbing
- 48. **Specialty:** Drainage
- 49. **Scenario:** Drain line needs to drop quickly around an obstruction.
- 50. **Problem:** Excess slope in a short section (e.g., 5 in drop over 4 ft) [21](#) [22](#) .
- 51. **Root Cause:** Lack of proper fittings to manage the vertical drop. [21](#) [22](#)
- 52. **Solution:** Use 45° or 22° fittings to make a wash-down point; angle down briefly then level out with another fitting [22](#) .
- 53. **Field Tip:** This method looks cleaner and reduces stress on the pipe [22](#) .
- 54. **Common Mistake:** Running drain pipe with continuous steep slope; may not align with trap or cause standing water [22](#) .
- 55. **Safety Note:** Glue fittings fully and allow cure time before running water.
- 56. **When Not to Use:** Long horizontal runs requiring constant slope.
- 57. **Confidence:** 5/10

58. **Tags:** plumbing, drainage, slope, fittings
59. **Source Link:** [21](#) [22](#)
60. **Supporting Quote:** "For a large drop, use 45s or even 22s... it makes a wash-down point and looks better" [22](#) .
61. **Maintain proper drain slope**
62. **Trade:** Plumbing
63. **Specialty:** Drainage
64. **Scenario:** Installing long drain lines.
65. **Problem:** Unsure if slope variations are acceptable.
66. **Root Cause:** Straying from recommended 1/4" to 1/2" per foot slope [21](#) .
67. **Solution:** Keep slope between 1/4" and 1/2" per foot; use fittings for sudden changes; avoid continuous steep drop [21](#) .
68. **Field Tip:** Use a torpedo level with built-in slope indicator to check runs.
69. **Common Mistake:** Letting a short section drop too quickly and causing traps to siphon.
70. **Safety Note:** Over-sloped drains can cause water to outrun solids; maintain code-compliant slope.
71. **When Not to Use:** Vertical stacks; these have their own rules.
72. **Confidence:** 5/10
73. **Tags:** plumbing, drain slope, code
74. **Source Link:** [21](#)
75. **Supporting Quote:** "Stick with a 1/4 to 1/2 in per foot slope" [21](#) .
76. **Know and use isolation valves**
77. **Trade:** Plumbing
78. **Specialty:** Maintenance
79. **Scenario:** Performing repairs on a fixture.
80. **Problem:** Shutting down water to the entire house for minor repairs.
81. **Root Cause:** Lack of isolation valves at individual fixtures [23](#) .
82. **Solution:** Install isolation valves for taps, toilets and other fixtures; they allow shutting off water locally for maintenance [23](#) .
83. **Field Tip:** Label your main shut-off and know how to operate it for emergencies [23](#) .
84. **Common Mistake:** Ignoring persistent leaks and delaying valve installation, leading to flooding during repairs.
85. **Safety Note:** Use proper pipe dope or thread tape when installing valves.
86. **When Not to Use:** Some jurisdictions require full-flow valves on certain fixtures; follow local codes.
87. **Confidence:** 5/10
88. **Tags:** plumbing, isolation valve, maintenance
89. **Source Link:** [23](#)
90. **Supporting Quote:** "Install isolation valves for individual fixtures... allow maintenance without shutting off the whole house" [23](#) .

91. **Avoid chemical drain cleaners**

92. **Trade:** Plumbing

93. **Specialty:** Maintenance

94. **Scenario:** Clearing slow or clogged drains.

95. **Problem:** Using aggressive chemicals damages plumbing.

96. **Root Cause:** Chemical cleaners eat away at pipes, wax rings, and rubber seals ²⁴ .

97. **Solution:** Use a plunger, drain snake or call a professional; avoid chemical cleaners ²⁴ .

98. **Field Tip:** Pour boiling water mixed with dish soap as a first step; it often removes grease.

99. **Common Mistake:** Repeatedly using chemicals which eventually ruin traps and fittings ²⁴ .

100. **Safety Note:** Chemical cleaners can cause burns; wear PPE and follow product instructions if used.

101. **When Not to Use:** Use only under guidance for septic systems; never mix with other chemicals.

102. **Confidence:** 5/10

103. **Tags:** plumbing, drain cleaners, safety

104. **Source Link:** ²⁴

105. **Supporting Quote:** "Putting drain cleaner down the toilet... it just eats up the wax ring" ²⁴ .

106. **Use proper fittings; avoid flexible drains**

107. **Trade:** Plumbing

108. **Specialty:** Drainage

109. **Scenario:** DIYers using accordion-style flexible drain lines.

110. **Problem:** Frequent clogs and leaks ²⁵ .

111. **Root Cause:** Flexible drain lines have ridges that collect debris and are not code compliant ²⁵ .

112. **Solution:** Use smooth, properly sized trap assemblies and fittings; avoid flexible accordion lines ²⁵ .

113. **Field Tip:** If a flexible line is necessary temporarily, replace it as soon as possible with rigid fittings.

114. **Common Mistake:** Installing flexible lines for convenience and leaving them permanently ²⁵ .

115. **Safety Note:** Use slip joint pliers gently to avoid cracking plastic fittings.

116. **When Not to Use:** High-temp drains; always choose approved materials.

117. **Confidence:** 4/10

118. **Tags:** plumbing, fittings, flexible line, code

119. **Source Link:** ²⁵

120. **Supporting Quote:** "Using flexible drain lines, the crinkled bendy ones, is a bad idea" ²⁵ .

Carpentry & Trim

1. **Protect your knees and pace yourself**

2. **Trade:** Carpentry

3. **Specialty:** Framing and general carpentry

4. **Scenario:** Long days on your knees laying out or installing flooring.

5. **Problem:** Knee pain and physical strain after decades in the trade.

6. **Root Cause:** Working on hard surfaces without padding and pushing too hard ²⁶ .

- 7. **Solution:** Use kneeling pads or knee pads; pace yourself; take breaks and stretch ²⁶ .
- 8. **Field Tip:** Becoming your own boss allows you to set a reasonable pace and schedule ²⁶ .
- 9. **Common Mistake:** Pushing hard every day and ignoring body signals; leads to injury ²⁶ .
- 10. **Safety Note:** Always wear knee protection when kneeling on concrete or decking.
- 11. **When Not to Use:** N/A—knee protection is always beneficial.
- 12. **Confidence:** 7/10
- 13. **Tags:** carpentry, safety, knee pads, pacing
- 14. **Source Link:** ²⁶
- 15. **Supporting Quote:** “Use kneeling pads and pace yourself... you don’t want to be broken later” ²⁶ .

16. Become your own boss to manage workload

- 17. **Trade:** Carpentry
- 18. **Specialty:** General
- 19. **Scenario:** Burnout from physically demanding carpentry work.
- 20. **Problem:** Long hours and heavy work take a toll on the body.
- 21. **Root Cause:** Working under bosses who push unrealistic productivity and overtime ²⁶ .
- 22. **Solution:** Start your own business or become a subcontractor to control pace and project selection ²⁶ .
- 23. **Field Tip:** Set boundaries and decline jobs that compromise your health or safety ²⁶ .
- 24. **Common Mistake:** Continuing to work for employers who overwork crews and neglect safety.
- 25. **Safety Note:** Being your own boss requires awareness of business risks and insurance.
- 26. **When Not to Use:** If you prefer steady employment or lack business experience.
- 27. **Confidence:** 5/10
- 28. **Tags:** carpentry, career, self-employment, pacing
- 29. **Source Link:** ²⁶
- 30. **Supporting Quote:** “Become your own boss... it’s easier on the body and you can set limits” ²⁶ .

31. Maintain core strength

- 32. **Trade:** Carpentry
- 33. **Specialty:** General physical maintenance
- 34. **Scenario:** Aging carpenter wants to stay in the trade without injury.
- 35. **Problem:** Lower back and core weakness leads to injury.
- 36. **Root Cause:** Neglecting core exercises ²⁷ .
- 37. **Solution:** Do Pilates, glute and core workouts regularly ²⁷ .
- 38. **Field Tip:** Incorporate stretching and strengthening into daily routine ²⁷ .
- 39. **Common Mistake:** Skipping exercise and focusing solely on work ²⁷ .
- 40. **Safety Note:** Consult a fitness professional if new to exercise.
- 41. **When Not to Use:** N/A—core strengthening benefits all trades.
- 42. **Confidence:** 6/10
- 43. **Tags:** carpentry, fitness, core strength, injury prevention
- 44. **Source Link:** ²⁷

45. **Supporting Quote:** "Pilates and glute/core workouts... keep me able to work" ²⁷ .

46. **Switch to finish carpentry or maintenance**

47. **Trade:** Carpentry

48. **Specialty:** Finish carpentry

49. **Scenario:** Aging carpenter wants to reduce physical strain.

50. **Problem:** Framing and concrete work are too physically demanding. ²⁸

51. **Root Cause:** Heavy lifting and repetition cause wear on joints and back.

52. **Solution:** Transition to finish carpentry, trim work, or maintenance roles; these are less physically taxing yet still pay well ²⁸ .

53. **Field Tip:** Develop fine woodworking skills and invest in quality finish tools ²⁸ .

54. **Common Mistake:** Sticking with heavy framing until injuries force retirement.

55. **Safety Note:** Protect eyes and lungs when cutting and sanding finish materials.

56. **When Not to Use:** If you enjoy heavy physical work and feel healthy.

57. **Confidence:** 6/10

58. **Tags:** carpentry, career path, finish work

59. **Source Link:** ²⁸

60. **Supporting Quote:** "Look into trim work or finish carpentry... you can make good money and it's much easier on the body" ²⁸ .

61. **Cope baseboard with a coping saw**

62. **Trade:** Carpentry

63. **Specialty:** Trim and finish

64. **Scenario:** Making inside corner joints on baseboard or crown molding.

65. **Problem:** 45° miters leave gaps when walls aren't perfectly square.

66. **Root Cause:** Miter joints open up due to out-of-square corners ²⁹ .

67. **Solution:** Cut a 45° miter on one piece; back cut along the profile with a coping saw at roughly 22.5° to remove waste; fit the coped piece against the adjacent molding ²⁹ .

68. **Field Tip:** Cut the coped piece a hair long and flex it into place for a tight fit ³⁰ .

69. **Common Mistake:** Trying to butt two 45° miter cuts together and expecting a perfect joint ²⁹ .

70. **Safety Note:** Use a sharp coping saw and clamp the workpiece securely.

71. **When Not to Use:** If molding is plastic or foam that doesn't cope well; then use a caulked miter.

72. **Confidence:** 8/10

73. **Tags:** carpentry, coping, baseboard, trim

74. **Source Link:** ²⁹ ³⁰

75. **Supporting Quote:** "Cut a 45°, then back cut at roughly 22.5° following the profile with a coping saw... cut a smidge long and snap it into place" ²⁹ ³⁰ .

76. **Use a jigsaw or grinder for coping**

77. **Trade:** Carpentry

78. **Specialty:** Trim and finish
79. **Scenario:** Coping crown molding or baseboard.
80. **Problem:** Manual coping saw is slow or difficult on hard woods.
81. **Root Cause:** Coping saw blades deflect or require high precision ³¹ ³² .
82. **Solution:** Use a jigsaw with a coping foot or an angle grinder with a flap disc to remove most of the waste; finish with a rasp or file for a clean profile ³¹ ³² .
83. **Field Tip:** Use an old circular saw blade to hog out the backside of crown molding before coping ³¹ .
84. **Common Mistake:** Trying to cope thick or dense material solely with a coping saw, leading to tear-out or broken blades.
85. **Safety Note:** Wear eye and hearing protection when using power tools; clamp molding securely. ³¹ ³²
86. **When Not to Use:** On soft pine or MDF where a coping saw is quicker.
87. **Confidence:** 6/10
88. **Tags:** carpentry, coping, tools
89. **Source Link:** ³¹ ³²
90. **Supporting Quote:** "I like using a jigsaw with a coping foot... some use an angle grinder with a flap wheel and then fine-tune with a file" ³¹ ³² .
91. **Snap long and flex into place**
92. **Trade:** Carpentry
93. **Specialty:** Trim and finish
94. **Scenario:** Installing baseboard or casing.
95. **Problem:** Slight gaps at corners when cut length exactly equals measurement.
96. **Root Cause:** Walls are not perfectly square or measurement has small error ³⁰ .
97. **Solution:** Cut the coped piece slightly long ($\frac{1}{16}$ "); flex it into place so that it springs tight into the corner ³⁰ .
98. **Field Tip:** Use a scrap piece to gauge the correct "spring back" length.
99. **Common Mistake:** Cutting exactly to measurement and ending up with gaps after the wood shrinks.
100. **Safety Note:** Be careful not to force the piece so hard that it snaps or bows.
101. **When Not to Use:** On brittle trim where flexing may cause cracks.
102. **Confidence:** 6/10
103. **Tags:** carpentry, trim, fitting
104. **Source Link:** ³⁰
105. **Supporting Quote:** "Cut a smidge long and snap it into place—it'll fit tight" ³⁰ .
106. **Trim waste on back of crown before coping**
107. **Trade:** Carpentry
108. **Specialty:** Crown molding
109. **Scenario:** Coping large or intricate crown profiles.
110. **Problem:** Cutting along the entire profile is tedious and prone to error ³¹ .
111. **Root Cause:** Too much material behind the face profile.

112. **Solution:** Use a circular saw or multi-tool to remove the majority of waste on the backside of the crown; this leaves only a thin profile to cope with a saw or file ³¹ .
113. **Field Tip:** Set the blade depth shallow to avoid cutting through the face of the molding. ³¹
114. **Common Mistake:** Attempting to cope the entire thickness, resulting in a wavy profile.
115. **Safety Note:** Secure the crown firmly and use clamps to prevent kickback.
116. **When Not to Use:** On small crown profiles where removal isn't necessary.
117. **Confidence:** 5/10
118. **Tags:** carpentry, crown molding, coping, waste removal
119. **Source Link:** ³¹
120. **Supporting Quote:** "Cut off the back of the crown with a circular saw and then cope... remove most of the material first" ³¹ .

HVAC/R

1. Clean coils and filters before diagnosis

2. **Trade:** HVAC/R

3. **Specialty:** Refrigerant troubleshooting

4. **Scenario:** High superheat with normal subcooling on a residential system.

5. **Problem:** Misdiagnosis due to dirty evaporator or condenser coils ³³ ³⁴ .

6. **Root Cause:** Heat transfer surfaces clogged with dirt and restricted airflow.

7. **Solution:** Clean the condenser coils thoroughly and replace or clean filters before checking superheat and subcooling ³³ ³⁴ .

8. **Field Tip:** Always start with a clean system; otherwise pressure readings are meaningless ³⁴ .

9. **Common Mistake:** Adding refrigerant to a system with dirty coils, causing overcharge and poor performance ³⁴ .

10. **Safety Note:** Turn off power and let coil cool before cleaning; use appropriate coil cleaner.

11. **When Not to Use:** When coil and filters are already known to be clean.

12. **Confidence:** 7/10

13. **Tags:** HVAC, superheat, subcooling, coil cleaning

14. **Source Link:** ³³ ³⁴

15. **Supporting Quote:** "Checking a system with dirty coils/filters is worthless; clean the condenser and change filters first" ³⁴ .

16. Identify TXV or restriction issues

17. **Trade:** HVAC/R

18. **Specialty:** Refrigerant metering devices

19. **Scenario:** High superheat with normal subcooling persists after cleaning coils.

20. **Problem:** Evaporator starved of refrigerant.

21. **Root Cause:** Bad TXV or liquid line restriction (filter-drier, kink) ³⁴ ³⁵ .

22. **Solution:** Test by warming or cooling the TXV sensing bulb; superheat should change; if not, replace the TXV; also check for restrictions and replace filter-drier ³⁵ .

23. **Field Tip:** Do not adjust TXV unless you confirm good airflow and refrigerant charge.

24. **Common Mistake:** Misdiagnosing piston metering devices as TXVs and replacing the wrong part ³⁶ .

25. **Safety Note:** Use proper protective equipment when handling refrigerant.

26. **When Not to Use:** When system uses a fixed orifice (piston)—apply other diagnostics.

27. **Confidence:** 6/10

28. **Tags:** HVAC, TXV, restriction, superheat

29. **Source Link:** ³⁴ ³⁵

30. **Supporting Quote:** “Warm or cool the TXV sensing bulb; if superheat doesn’t change, the TXV is bad” ³⁵ .

31. Distinguish piston vs TXV systems

32. **Trade:** HVAC/R

33. **Specialty:** Refrigerant diagnosis

34. **Scenario:** Identifying cause of high superheat on a system.

35. **Problem:** Misdiagnosis of metering devices.

36. **Root Cause:** Not knowing whether the system uses a fixed orifice (piston) or TXV ³⁶ .

37. **Solution:** Check the equipment documentation or visually inspect the evaporator; low superheat with low subcooling typically indicates undercharge on piston systems; high superheat with normal subcooling points to restriction or TXV issue ³⁶ .

38. **Field Tip:** Always inspect the type of metering device before adding or removing refrigerant. ³⁶

39. **Common Mistake:** Replacing TXV on a piston system, wasting time and money ³⁶ .

40. **Safety Note:** Ensure power is off when inspecting air handler components.

41. **When Not to Use:** When the system uses electronic expansion valves (EEVs)—follow manufacturer procedure.

42. **Confidence:** 5/10

43. **Tags:** HVAC, piston, TXV, diagnosis

44. **Source Link:** ³⁶

45. **Supporting Quote:** “A piston will give low superheat and low subcooling when undercharged, but a TXV will present differently” ³⁶ .

46. Check return air ducts and airflow

47. **Trade:** HVAC/R

48. **Specialty:** Airflow diagnosis

49. **Scenario:** High superheat persists after cleaning coils and verifying TXV.

50. **Problem:** Starved evaporator due to insufficient airflow.

51. **Root Cause:** Blocked return air ducts or undersized ductwork ³⁶ .

52. **Solution:** Inspect and clean return air ducts; measure static pressure; ensure duct sizing is adequate; if necessary, modify or add returns ³⁶ .

53. **Field Tip:** High superheat with high suction pressure can indicate low airflow; check blower speed settings.

54. **Common Mistake:** Focusing solely on refrigerant charge without verifying airflow.

55. **Safety Note:** Always secure panels and disconnect power before inspecting ducts.

56. **When Not to Use:** When airflow has been verified with a manometer.
57. **Confidence:** 5/10
58. **Tags:** HVAC, airflow, diagnosis, ductwork
59. **Source Link:** ³⁶
60. **Supporting Quote:** "Check return ducts... high superheat with normal subcool could be a return air issue" ³⁶ .
61. **Hire reputable HVAC professionals**
62. **Trade:** HVAC/R
63. **Specialty:** Service selection
64. **Scenario:** Homeowner frustrated by company insisting there is no problem.
65. **Problem:** Poor diagnosis and lack of clear explanation from service company ³⁶ .
66. **Root Cause:** Hiring unqualified or disinterested technicians.
67. **Solution:** Seek companies with good reviews and thorough diagnostic processes; ask them to explain what they found and why ³⁶ .
68. **Field Tip:** Request that service techs provide static pressure readings and superheat/subcool numbers as proof of diagnosis.
69. **Common Mistake:** Accepting the answer "there's no problem" when symptoms clearly suggest otherwise. ³⁶
70. **Safety Note:** Unqualified work can void warranties; choose certified professionals.
71. **When Not to Use:** If you are comfortable performing basic diagnostics yourself.
72. **Confidence:** 4/10
73. **Tags:** HVAC, service, professional
74. **Source Link:** ³⁶
75. **Supporting Quote:** "Find a good company... call someone who will test the system properly instead of saying there's no issue" ³⁶ .

Pipefitting / Ironwork

1. **Leave clearance for insulation**
2. **Trade:** Pipefitting
3. **Specialty:** Spool fabrication
4. **Scenario:** Prefabricated spool fit-up in a plant.
5. **Problem:** After installation, there is no room for insulation around the pipe ³⁷ .
6. **Root Cause:** Fitter measured from centreline to centreline without adding clearance for insulation or jacketing ³⁷ .
7. **Solution:** Add insulation thickness to spool measurements; ensure enough gap between parallel runs to allow insulation ³⁷ .
8. **Field Tip:** Consult the insulation spec early; pre-wrap test pieces or measure jacketing thickness ³⁷ .
9. **Common Mistake:** Forgetting to account for insulation, leading to rework and schedule delays ³⁷ .
10. **Safety Note:** Hot pipes require proper insulation thickness to prevent burns.
11. **When Not to Use:** Pipes that are not insulated (e.g., instrument air lines).

12. **Confidence:** 6/10
13. **Tags:** pipefitting, spool, insulation clearance
14. **Source Link:** ³⁷
15. **Supporting Quote:** "Always leave enough room for insulation... older insulators emphasise this" ³⁷.
16. **Stagger tie-ins and check square**
17. **Trade:** Pipefitting / Welding
18. **Specialty:** Spool installation
19. **Scenario:** Fitting multiple tie-ins on a spool with TIG terminations.
20. **Problem:** Misaligned tie-in points and poor fit-up ³⁸.
21. **Root Cause:** Placing tie-ins too close together and failing to check squareness ³⁸.
22. **Solution:** Stagger and overlap tie-in points; run out TIG terminations smoothly; always check that spool is square after tacking ³⁸.
23. **Field Tip:** Use a level and two squares; measure diagonals to ensure squareness before final weld ³⁸.
24. **Common Mistake:** Tack-welding without checking square, leading to misalignment ³⁸.
25. **Safety Note:** Secure spools properly when handling; heavy spools can shift.
26. **When Not to Use:** On single-tie-in spools where alignment is straightforward.
27. **Confidence:** 5/10
28. **Tags:** pipefitting, spool, tie-in, squareness
29. **Source Link:** ³⁸
30. **Supporting Quote:** "Stagger and overlap your tie-in points... run out TIG terminations and double check square after tacking" ³⁸.

Professional Development & Misc

1. **Continue training and embrace new technologies**
2. **Trade:** Electrical (general trades advice)
3. **Specialty:** Career development
4. **Scenario:** New journeyman entering the field.
5. **Problem:** Falling behind on technology and code changes.
6. **Root Cause:** Not dedicating time to education and training ³⁹.
7. **Solution:** Take classes, obtain certifications, and stay current with codes and emerging technologies ³⁹.
8. **Field Tip:** Join professional organisations and learn from inspectors and seasoned electricians ³⁹.
9. **Common Mistake:** Thinking you know everything after apprenticeship; failing to keep up with code changes ³⁹.
10. **Safety Note:** Updated knowledge prevents mistakes that could cause fires or shock.
11. **When Not to Use:** Never stop learning; always applicable.
12. **Confidence:** 6/10
13. **Tags:** professional development, training, electrical
14. **Source Link:** ³⁹

15. **Supporting Quote:** “Learn new technologies; estimate task durations; delegate to apprentices and communicate clearly” 39 .

16. **Estimate duration and delegate**

17. **Trade:** Electrical / general trades

18. **Specialty:** Job management

19. **Scenario:** Journeyman running a job site with apprentices.

20. **Problem:** Underestimating time and not delegating properly.

21. **Root Cause:** Inexperience in planning and communication 39 .

22. **Solution:** Estimate how long tasks will take, plan breaks and milestones; delegate simpler tasks to apprentices to stay on schedule 39 .

23. **Field Tip:** Communicate openly with apprentices about expectations and safety; ask inspectors if uncertain 39 .

24. **Common Mistake:** Trying to do everything yourself and falling behind schedule 39 .

25. **Safety Note:** Supervising trainees reduces errors and accidents.

26. **When Not to Use:** Very small one-person jobs.

27. **Confidence:** 5/10

28. **Tags:** job management, delegation, planning, electrical

29. **Source Link:** 39

30. **Supporting Quote:** “Estimate how long things should take and delegate; talk to your apprentices, inspector and homeowner” 39 .

31. **Communicate with inspectors and clients**

32. **Trade:** Electrical / general trades

33. **Specialty:** Professional communication

34. **Scenario:** Working on projects that require permits and inspections.

35. **Problem:** Misunderstandings with inspectors or clients leading to rework 39 .

36. **Root Cause:** Poor communication and not asking questions.

37. **Solution:** Develop a rapport with inspectors; ask questions when uncertain; explain your plan to homeowners 39 .

38. **Field Tip:** Keep a notebook of inspector comments for future reference 39 .

39. **Common Mistake:** Avoiding inspectors or failing to clarify requirements 39 .

40. **Safety Note:** Good communication ensures code compliance and safety.

41. **When Not to Use:** N/A.

42. **Confidence:** 5/10

43. **Tags:** communication, inspector, client

44. **Source Link:** 39

45. **Supporting Quote:** “Don’t be afraid to ask your inspector or homeowner questions—communication goes a long way” 39 .

46. **Hydrate and eat well**

47. **Trade:** Carpentry / general trades

48. **Specialty:** Health

49. **Scenario:** Tradespeople working long days in hot or cold environments.

50. **Problem:** Fatigue, cramps, and reduced performance due to dehydration and poor diet ⁴⁰ .

51. **Root Cause:** Neglecting water intake and nutrition. ⁴⁰

52. **Solution:** Drink water throughout the day, even when not thirsty; pack balanced meals and snacks ⁴⁰ .

53. **Field Tip:** Limit sugary drinks and coffee; they dehydrate. ⁴⁰

54. **Common Mistake:** Skipping lunch or drinking only energy drinks. ⁴⁰

55. **Safety Note:** Dehydration can lead to heat exhaustion or fainting.

56. **When Not to Use:** N/A—hydration always matters.

57. **Confidence:** 4/10

58. **Tags:** health, hydration, trades

59. **Source Link:** ⁴⁰

60. **Supporting Quote:** “Diet and hydration are huge... don’t be afraid to tell the boss you’re going to see your kid’s soccer” ⁴⁰ .

61. Use kneeling pads and pacing

62. **Trade:** Carpentry / general trades

63. **Specialty:** Health & pacing

64. **Scenario:** Tradesperson working on knees for extended periods.

65. **Problem:** Joint pain and injury ²⁶ .

66. **Root Cause:** Hard surfaces and constant kneeling without protection ²⁶ .

67. **Solution:** Use high-quality kneeling pads; take breaks to stretch; alternate tasks to reduce repetitive stress ²⁶ .

68. **Field Tip:** Invest in gel knee pads or foam pads; replace them when they compress. ²⁶

69. **Common Mistake:** Working through pain and refusing pads because “tough guys don’t use knee pads.” ²⁶

70. **Safety Note:** Chronic knee damage can shorten a career.

71. **When Not to Use:** N/A.

72. **Confidence:** 7/10

73. **Tags:** health, knee pads, pacing, carpentry

74. **Source Link:** ²⁶

75. **Supporting Quote:** “Use kneeling pads and pace yourself or you’ll be broken later” ²⁶ .

76. Do Pilates for longevity

77. **Trade:** Carpentry / general trades

78. **Specialty:** Fitness

79. **Scenario:** Tradesperson wanting to remain limber and strong.

80. **Problem:** Muscles tighten and joints stiffen over years of work ²⁷ .

81. **Root Cause:** Repetitive movements and lack of stretching ²⁷ .
82. **Solution:** Incorporate Pilates and glute/core exercises into routine; improves flexibility and core strength ²⁷ .
83. **Field Tip:** Do 10-minute stretching every morning before work ²⁷ .
84. **Common Mistake:** Avoiding exercise due to lack of time or interest. ²⁷
85. **Safety Note:** Warm up properly; consult doctor if new to exercise.
86. **When Not to Use:** N/A.
87. **Confidence:** 5/10
88. **Tags:** health, fitness, Pilates, trades
89. **Source Link:** ²⁷
90. **Supporting Quote:** "Pilates and glute/core workouts keep me able to work" ²⁷ .
91. **Learning from mistakes and shifting roles**
92. **Trade:** Carpentry / general trades
93. **Specialty:** Career progression
94. **Scenario:** Carpenter transitions from lead carpenter to superintendent. ⁴¹
95. **Problem:** Physical work becomes harder with age, but career satisfaction is still desired.
96. **Root Cause:** Doing heavy lifting for decades leads to burnout. ⁴¹
97. **Solution:** Move into supervisory or superintendent roles; still stay connected to the trade but with less physical strain ⁴¹ .
98. **Field Tip:** Even as superintendent, keep your hands in the work occasionally to stay sharp; enjoy training others.
99. **Common Mistake:** Waiting until injuries force a role change instead of planning ahead.
100. **Safety Note:** Supervisory roles come with mental stress; develop stress-management techniques.
101. **When Not to Use:** If you enjoy hands-on work and your body allows it.
102. **Confidence:** 4/10
103. **Tags:** career progression, carpentry, superintendent
104. **Source Link:** ⁴¹
105. **Supporting Quote:** "I moved from lead carpenter to superintendent... still set doors and trim but it's less physical" ⁴¹ .

¹ ² 6013 Slag inclusion : Welding

https://old.reddit.com/r/Welding/comments/bza3y1/6013_slag_inclusion/

³ ⁴ Any tips for 7018 restarts? : Welding

https://old.reddit.com/r/Welding/comments/11xths3/any_tips_for_7018_restarts/

⁵ ⁶ ⁷ Help fix vertical up welds : r/Welding

https://www.reddit.com/r/Welding/comments/18ed6qz/help_fix_vertical_up_welds/

⁸ ⁹ ¹⁰ ¹¹ ¹² ¹³ Getting bad TIG welds randomly. I changed gas tank, tungsten, and filler but still happens. Acting like metal is contaminated but I don't think it is. Looks like a sparkler on the 4th of July and I end up with welds looking like this. Do I have a leak in my whip? : r/Welding

https://www.reddit.com/r/Welding/comments/12pkpd6/getting_bad_tig_welds_randomly_i_changed_gas_tank/

14 15 Struggling with Offsets in Conduit Bending : r/electricians

https://www.reddit.com/r/electricians/comments/1j8104o/struggling_with_offsets_in_conduit_bending/

16 17 18 How to fix leak in new compression fitting : r/fixit

https://www.reddit.com/r/fixit/comments/1h7wk63/how_to_fix_leak_in_new_compression_fitting/

19 20 PEX leak by circulation pump. : r/Plumbing

https://www.reddit.com/r/Plumbing/comments/16kip2g/pex_leak_by_circulation_pump/

21 22 23 24 25 Drain slope : r/Plumbing

https://www.reddit.com/r/Plumbing/comments/1gpob6d/drain_slope/

26 40 41 Carpenters over the age of 30 - How do you find the energy for things in your life other than work? : Carpentry

https://old.reddit.com/r/Carpentry/comments/1d0mmxv/carpenters_over_the_age_of_30_how_do_you_find_the/

27 28 Older Carpenters who've taken care of themselves, how do you feel? : r/Carpentry

https://www.reddit.com/r/Carpentry/comments/1pjb5a4/older_carpenters_whove_taken_care_of_themselves/

29 30 31 32 Best way to cope trim? : r/Carpentry

https://www.reddit.com/r/Carpentry/comments/1dp6a96/best_way_to_cope_trim/

33 34 35 36 High Superheat but Normal Subcooling? : r/hvacadvice

https://www.reddit.com/r/hvacadvice/comments/1nzi41z/high_superheat_but_normal_subcooling/

37 38 Give me some old tips/ tricks for fitting up pipe. I'll take any advice : r/pipefitter

https://www.reddit.com/r/pipefitter/comments/1jw7usm/give_me_some_old_tips_tricks_for_fitting_up_pipe/

39 Experienced journeymen/foremen/supers/owners, what advise would you give to a newly minted journeyman? : electricians

https://old.reddit.com/r/electricians/comments/1u0m6xy/experienced_journeymenforemensupersowners_what/